Understanding Student Mathematics Achievement in Diverse Socio-economic Contexts: A Multi-level Analysis of Private Schools in the United Arab Emirates

Abstract

This study investigates the multi-faceted factors that influence student mathematics achievement using the Programme for International Student Assessment 2022 dataset, offering insights into the under-researched private schools of the United Arab Emirates, a high-performing urban educational system with migrant student population of substantial socio-economic differences. Multi-level analysis was employed to examine how student characteristics and attitudes, home environments, and school-level variables converge to shape student mathematical outcomes. Among the most significant predictors emerged were math self-efficacy, immigrant status, math anxiety, and the school level socio-economic status. The negative effects of social relationships, well-being, and math motivation on math achievement were also revealed and interpreted in the context of the UAE's private schools. It is argued that the UAE private schools' high average performance in PISA is driven by a combination of high-performing school segments and a relatively advantaged immigrant student population, but internal inequities by school quality and socio-economic background remain.

Keywords: math achievement; multi-level modelling; self-efficacy; socio-economic status

Introduction

Improving students' mathematics achievement has been a major priority in education systems worldwide. Mathematics proficiency not only predicts individuals' future academic trajectories and labour-market opportunities (Rose, 2006), but also constitutes a core competency required

in the twenty-first century. As the foundation of modern science and technology, mathematics plays a crucial role in supporting technological advancement, expansion of digital economies, and economic development in general (OECD, 2023a).

Research has highlighted a wide range of determinants of student mathematics achievement, including mathematics anxiety, mathematics self-efficacy, instructional quality, and family socio-economic status (SES) (Breit et al., 2025; Güzeller et al. 2016; Wang et al., 2023). Despite the growing literature, the complexity of educational dynamics in different educational and social contexts often leads to contradictory or counterintuitive findings (Erdogdu & Erdogdu, 2025). Significant gaps remain in our understanding of what affect students' math learning and performance, which tends to context dependent (Wang et al., 2023). For instance, strong evidence shows anxiety correlates negatively with performance, but fewer studies test the potential mechanisms in authentic classrooms and school settings (Ashcraft & Krause, 2007). Moreover, disparities in mathematics achievement are not merely indicators of academic performance; they are also closely linked to broader patterns of social inequality (Campos et al., 2025; Coleman et al., 1966; OECD, 2023b). Therefore, analyses focusing solely on individual-level attributes may overlook the broader contextual factors that influence educational outcomes. A more comprehensive understanding of mathematics achievement requires examining how individual and contextual factors interact within educational systems.

International large-scale assessments, such as the Programme for International Student Assessment (PISA), conceptualize student achievement as the outcome of interactions among student-, home-, classroom-, school-, and system-level factors. This perspective highlights the importance of adopting multi-level analytical approaches when studying educational outcomes.

Mathematics was the major domain assessed in PISA 2022 and remains a key priority within the education reform agenda of the United Arab Emirates (UAE). Despite significant investments in education, the UAE's average mathematics score in PISA remains below the OECD average, with a mean score of 431 in 2022 (OECD, 2023b). Considerable variation exists within the UAE's education system. Notably, students in private schools significantly outperform those in public schools. In PISA 2022, private schools in the Emirate of Dubai achieved an average mathematics score of 497, positioning them among the highest-performing subnational education systems participating in PISA (KHDA, 2023). The UAE's private school sector represents a high-performing urban education system embedded within a rapidly developing national context and characterized by substantial diversity in curricula, assessment, student socio-economic backgrounds, immigrant status, and cultural norms. This context provides a valuable opportunity to examine how student-level and school-level factors interact to influence mathematics learning and achievement in an under-researched educational setting.

In the literature, relatively little evidence exists in the Gulf Cooperative Council (GCC) countries such as the UAE that have migrant-majority educational systems, where socio-economic and cultural mechanisms may operate differently from the Western contexts. Findings from the UAE PISA 2022 indicate that socio-economic status and other background characteristics exert a strong influence on student performance, while gender gaps in mathematics tend to favour girls (OECD, 2023c). However, few studies have employed multi-level analytical approaches to examine how student- and school-level factors jointly influence mathematics achievement in the UAE. For example, using PISA 2018 data, Marquez et al. (2022) applied linear regression models to identify achievement gaps across emirates, nationality groups, gender, and school types, but did not account for cross-level interactions. Badri et al. (2020) analyzed TIMSS 2015 data and found that individual level variables such as

students' attitudes toward mathematics and their perceptions of the relevance of mathematics to future life were among the strongest predictors of mathematics performance. It was not clear, however, whether plausible values were incorporated in the structural equation modelling. More recently, Winchip (2024) highlighted achievement disparities between immigrant and non-immigrant students in the UAE, employing thematic analysis of PISA 2022. Thematic analysis may overlook the compounding effects among variables. While these studies provide valuable insights, more rigorous analyses using appropriate multi-level modelling techniques are needed to better capture the hierarchical structure of international large-scale assessment data.

This study aims to investigate the determinants of math achievement among students attending private schools in the UAE by applying multi-level modelling to PISA 2022 data. It focuses on the interactions between student characteristics and school environments, and explores how home and school contexts interact with students' cognitive and motivational attributes. Insights from high-performing private schools may provide useful implications for policy and practice.

Literature Review: Determinants of Student Mathematics Achievement

Among the theoretical perspectives that explain how students develop mathematical competence, socio-cultural approaches stress the importance of social interaction, cultural and language, and contextual factors in shaping learning processes (Vygotsky, 1978). Learning occurs through participation in socially organized practices, where norms and interactions influence students' engagement and performance (Cobb & Yackel, 1996). Expanding this perspective, the social-ecological systems theory proposes that students' performance are influenced by multiple nested environmental systems, including family, school, policies, and the broader societal context (Bronfenbrenner, 1981). It should be noted that the design and

methodology of PISA tend to align with Vygotsky's (1978) social constructivist theory and the socio-ecological theory, which together provide a framework for the literature review and analysis of this study. Among various factors that may influence student maths performance, this section focuses on those are available in PISA data and groups them into four categories: student characteristics and attitudes, family and home background, teacher and instructional factors, and school or system-level conditions.

Among student characteristics and attitudes, early literacy and numeracy abilities consistently emerge as strong predictors of subsequent academic performance across subject areas, including mathematics (Breit et al., 2025; Chang, 2023). Beyond cognitive ability, students' attitudes toward mathematics, including self-efficacy, enjoyment, and anxiety, contribute additional explanatory power (Hoffman, 2010; Jansen et al, 2016). Analyses of PISA data demonstrate that mathematics self-efficacy in particular significantly predict mathematics performance in various settings (Williams & Williams, 2010). Systematic reviews of the self-efficacy-achievement relationship likewise report positive associations, although variations may occur across demographic groups and educational contexts (Huang, 2013). Longitudinal research further suggests that self-efficacy not only predicts later mathematics performance but also mediates the relationship between motivation, anxiety, and academic achievement (Schöber et al., 2018). It is worth noticing that improvement in self-efficacy may have varying level of effects on student achievement across race and gender (Cheema & Kitsantas, 2014).

Research evidence indicates a small to moderate negative relationship between mathematics anxiety and achievement (Zhang et al., 2019). For instance, Xiao and Sun (2021) found that American students with lower levels of anxiety and higher motivation achieved higher scores. Similarly, Chan and Liem (2023) identified a negative association between mathematics

anxiety and achievement among secondary school students in Singapore. Mathematics confidence or anxiety may result from school climate and social connection. While school-related factors such as high academic demands and frequent testing may further contribute to stress and anxiety among students (Högberg et al., 2019), effective teacher-student interactions could promote motivation in mathematics among students (Durksen et al., 2017).

Cognitive and motivational factors often interact with gender norms and broader cultural expectations. In some educational systems, female students report higher levels of mathematics anxiety despite achieving equal or higher performance compared to male students (Justicia-Galiano et al., 2023). Analyses of TIMSS 2019 data across 39 countries revealed that 8th grade boys generally reported higher levels of mathematics confidence than girls (Campos et al., 2025). Differences in self-efficacy and motivation, often reported to be lower among female students, have been identified as potential explanations for gender disparities in mathematics achievement (Rodríguez et al., 2020). Other studies have also found gender differences favouring boys in mathematics performance, attributing these patterns partly to socio-cultural and economic factors (Molina-Muñoz et al., 2023). The gender gaps in mathematics performance among Malaysian students disappeared when socio-economic status at student and school level was accounted for (Thien, 2016). Several studies revealed a significant moderating role of gender egalitarian values in gender differences in adolescents' academic achievement (Gevrek et al., 2020). Nevertheless, Anghel et al.'s (2020) across- and within-countries analysis using five waves of PISA data suggested that the positive association between gender gap in math scores and country-level gender equality disappears in OECD countries, suggesting the interplay of country level economic factors.

SES is among the most robust predictors of mathematics achievement. Students from socio-economically more advantaged families tend to achieve higher mathematics scores, because they often have greater access to educational resources, supportive learning environments, and opportunities that foster cognitive and socio-emotional development (James-Brabham et al., 2023; Perry et al., 2022). Systematic reviews focusing on studies using PISA data identify family socio-economic background as a consistent predictor of achievement (Wang et al., 2023), which is congruent with broader international research findings (Xie & Ma, 2019). However, analyses of PISA data suggest that not all components of SES contribute equally to mathematics achievement, indicating the need for careful consideration of different SES indicators (Lee & Borgonovi, 2022). As suggested in van Ewijk and Sleegers' (2010) meta-analysis, the size of compositional effects of SES tends to vary depending on how SES is measured and whether prior achievement is controlled for.

A comprehensive earlier review by Desforges and Abouchaar (2003) concluded that parental involvement, particularly parents' educational aspirations for their children and their support for learning activities, can positively influence students' academic achievement and learning behaviours. Subsequent studies have associated parental involvement with improved school attendance, more positive student attitudes toward learning, and higher assessment scores (El Nokali et al., 2010). Research conducted among adolescents in Chile and Ecuador found that parental behaviours such as monitoring and supportive guidance positively influenced students' achievement orientation and self-efficacy (Ingoldsby et al., 2003). Importantly, the nature of parental involvement also matters. Supportive engagement may promote motivation and achievement, whereas controlling or punitive practices may contribute to increased anxiety and lower motivation (Gordeeva et al., 2024). Nevertheless, several barriers may limit parental engagement, particularly among families from lower socio-economic backgrounds. These

barriers include limited cultural capital, language challenges, and difficulties in establishing effective relationships with schools (Levine-Rasky, 2009; Skaliotis, 2010).

Meta-analytic research on instructional effectiveness suggests that structured and cognitively challenging instruction is associated with improved academic achievement (Kyriakides et al., 2013). Recent studies further support the importance of instructional practices. For example, Elouafi et al. (2026) applied machine-learning techniques to identify contextual factors influencing Moroccan students' mathematics performance and found that pedagogical practices were among the strongest predictors of achievement. Specific instructional strategies, such as tutoring, cooperative learning, and well-structured teaching approaches, have demonstrated positive effects on mathematics achievement, although their impact varies depending on grade level and implementation quality (Barbieri et al., 2023). Across education systems, the association between teaching strategies and mathematics performance of students interacts with the socio-economic and instructional context (Caro et al., 2016).

Teacher expectation and relationship with students represent another influential factor in shaping student performance. Research suggests that teachers' expectations can affect student motivation, engagement, and achievement through mechanisms such as self-fulfilling prophecies (de Boer et al., 2018). Analysing PISA 2012 and 2022 data from ten European countries, Liu et al. (2024) found that teacher support and a positive disciplinary climate were associated with improved student performance, particularly among students from lower socio-economic backgrounds. Other instructional quality and classroom environment related variables such as cognitive activation, classroom management, and student-oriented instruction showed mixed relationships with achievement outcomes.

School-level characteristics, including teacher qualifications, leadership practices, resource availability, and school climate, contribute to differences in student learning outcomes (Berkowitz, 2020; Wang & Degol, 2016; Wang et al., 2023). In particular, research indicates that differences in students' mathematical competence can partly be explained by the socio-economic composition of the schools (Boda et al., 2022). For example, analyses of PISA data from 27 countries found that school SES had a positive effect on students' reading performance (León et al., 2022). This is confirmed by the analysis of PISA 2018 Ireland data in reading and mathematics (Flannery et al., 2023). Both OECD's (2010) analysis and meta-analysis (Sirin, 2005) indicated a larger level of association between math achievement and school SES than between math achievement and individual SES.

At a broader level, institutional arrangements within education systems may reinforce or mitigate socioeconomic inequalities in student learning outcomes (Le Donné, 2014). System-level policies, including curriculum design, teacher recruitment and training, accountability frameworks, and resource allocation, shape the conditions under which schools operate (Ghamrawi, 2023). Recent research using PISA 2022 data highlights the importance of educational resources and school climate in influencing student achievement. Erdogdu and Erdogdu (2025) reported that access to educational resources had a strong positive effect on student performance, while adverse school climates were associated with lower achievement.

Examining PISA mathematics scores across 59 countries, Breton (2023) attributed national culture as the key underlying determinant of student achievement, as it explained 62% of the variation in the average scores. Across societies and cultures, cultural value is positively associated with student academic results (King et al., 2020). Furthermore, various studies have shown that the relationship between social and cultural context and achievement is mediated by

school or classroom factors, such as teacher expectations, quality of instruction, or adequacy of school resources (Liu et al., 2015; Opdenakker & van Damme, 2001).

Taken together, while the literature consistently identifies several key determinants of mathematics achievement, including prior knowledge, self-efficacy, mathematics anxiety, SES, and instructional quality, evidence on the effects of other factors is rather mixed and important limitations remain (Wang et al., 2023). More research is needed to examine heterogeneous effects across different student groups, such as variations related to SES, language background, migration status, and gender, in different social-economic systems and educational settings, especially in analyses using large-scale international assessment data (Liu et al., 2024). Moreover, variables at different levels should not be analyzed in isolation, as both socio-ecological theory and empirical studies reviewed above suggest that mathematics achievement is shaped by a multi-layered set of influences.

Therefore, employing multi-level modelling to PISA 2022 UAE private school student data, this study aims to answer the following research questions: 1) What are the most significant determinants of math achievement for students attending private schools in the UAE? 2) Which level of variables have larger influence on math achievement: individual level or school level? 3) Does the relationship between student SES and achievement vary across schools? 4) To what extent socio-economic differences in math scores are moderated by immigrant background, parent involvement, and school socioeconomic composition? and 5) How students' motivational differences in math scores are moderated by school contexts?

Data and Methods

Sample

In mathematics, PISA assesses 15-year-olds' capacity to reason mathematically and to formulate, employ, and interpret mathematics to solve problems in a variety of real-world contexts. Thus, mathematics achievements reported in PISA present suitable data to explore the influence of learning environment on student achievement.

A robust two-stage sampling procedure was followed in PISA to select students that allowed better representation and consequently greater data generalizability. In stage 1, schools were sampled with probability proportional to size within certain strata. In stage 2, a number of 15-year-old students were selected at random within each sampled school. As a result, the UAE's PISA 2022 sample comprised 24,600 students from 840 schools and representing the population of 15-year-olds in the UAE.

This study focuses on the UAE's private schools that are known for their diversity in curricula, learning environment, school cultural norms, and student population in terms of socio-economic and ethnical backgrounds, as against public schools that are predominantly attended by Emirati students (Garces-Bacsal et al., 2024; Hefnawi, 2026; OECD, 2021). For the analysis, the student dataset was merged with the school questionnaire. After the merging of datasets, schools without school questionnaire data were removed; schools with less than 10 students participating were removed; cases with substantial missing values, especially in key variables, were also excluded. After data cleaning and arrangement, 10,668 cases of data from 104 private schools were processed (Table 1).

(Table 1 about here)

As presented in Table 1, the sample of UAE private school students participated in PISA 2022 is gender-balanced, featuring a high proportion (47.6%) of 1st generation immigrant students and a

relatively socio-economically advantaged student population. It should be noted that 2nd generation immigrant students in the context of the UAE may differ from Emirati students in language, socioeconomic resources, and educational expectations. There are also indications that considerable variations exist among private schools in terms of proportion of teacher-certification and teacher-student ratio. These make the sample suitable for the examination of cross-level effects of socio-economic and immigration backgrounds on math achievement.

Variables

In PISA, mathematics achievement is reported through plausible values, multiple imputed draws from the posterior distribution of proficiency for each student. The dependent variable used in the following multi-level analysis is student mathematics achievement, measured via PISA mathematics plausible values. PISA designed and combined many questionnaire items to represent latent constructs and scaling procedures were applied to create weighted likelihood estimates, which were standardised (OECD, 2024). Informed by the relevant literature reviewed above and based on the data arranged by PISA, the following student-level predictors were selected in the analysis.

Student characteristics and math attitudes variables included: gender (female = 1, male = 0); immigrant background (IMMIG) (first generation = 1, native and second-generation = 0); preference of math over other core subjects (MATHPREF) (yes = 1, no = 0); studying for school or homework before or after school (STUDYHMW) (not every day = 1, no = every day or more); motivation to do well in mathematics (MATHMOT) (more motivated = 1, not more motivated = 0); mathematics anxiety (ANXMAT); curiosity (CURIOAGR); and mathematics self-efficacy (MATHEFF). The other mathematics self-efficacy variable was dropped due to collinearity concerns. Several safety and psychological outcomes variables were also selected,

including being bullied (BULLIED), feeling safe (FEELSAFE), and students life satisfaction across domains (LIFESAT).

Math teaching and learning related variables reported by students included quality of student-teacher relationships (RELATST); disciplinary climate in mathematics (DISCLIM); cognitive activation in mathematics (COGACMCO); and exposure to formal and applied mathematics tasks (EXPOFA). Home context variables included at the student-level are family support (FAMSUP); social connection to parent (SOCONPA); social connections (SOCCON); ICT resources (ICTRES); and economic, social, and cultural status (ESCS). In PISA, ESCS is an index variable constructed from students' responses to questions about the highest level of education of their parents, parental occupation, and amount of home possessions. Computed for international comparability, ESCS represents the operationalization of SES.

School-level predictors included proportion of all teachers fully certified (PROATCE); teacher participation (TCHPART); school encouragement of parent involvement (ENCOURPG); instructional leadership (INSTLEAD); math teacher training (MTTRAIN); negative school climate (NEGSCLIM); teacher-related factors affecting school climate (TEACHBEHA); mathematics-related extra-curricular activities at school (MACTIV) (0 or 1 math ECA = 1, 2 or more math ECA = 0); ability grouping for mathematics classes (ABGMATH) (no grouping = 1, yes = 0); and school mean ESCS (Mean_ESCS).

Several interaction variables were included to reveal potential multi-level moderation effects: ANXMAT x NEGSCLIM; ANXMAT x Mean_ESCS; IMMIG x NEGSCLIM; ESCS x ENCOURPG; and ESCS x Mean_ESCS. ESCS x IMMIG was also added to examine the interactional effects between two individual-level variables on math achievement.

Patterns of missing values among the selected variables were assessed. The percentage of missing values was below 5%. However, Little's MCAR test was significant (p = .000), indicating that the missingness was not completely at random. Multiple imputation was used to handle the missing values. Prior to running multi-level regression, preliminary analyses were also conducted to test the assumptions of homoscedasticity and multi-collinearity. A scatterplot of standardized residuals versus predicted values was produced and inspected. Given that the residual plot was in a roughly rectangular shape with points around the horizontal zero line and with the same vertical spread, the regression model did not appear to suffer from heteroscedasticity. Inter-variable correlations were generated to detect potential multi-collinearity. The correlation coefficients between the independent variables in the model were all well below the threshold of 0.7 (Tabachnick & Fidell, 2007). Due to space reason, Table 2 only contains correlation coefficients between independent variables reported by students, which range from 0.001 to 0.408. Collinearity among the variables were also checked through examining the Variance Inflation Factor (VIF). The small values of VIF, ranging from 1.047 to 4.153, confirmed that there is no problem of collinearity (Knoke et al., 2002).

(Table 2 about here)

Statistical Modelling

Given that students are nested within schools, multi-level modelling was used to account for clustering and to examine both student- and school-level influences on mathematics achievement (Raudenbush & Bryk, 2002). Specifically, a two-level hierarchical linear model was conducted using MLwiN (version 2.30) to analyse the UAE private school students mathematics achievements in PISA 2022.

Five models were fitted. A baseline two-level random intercept model was specified as:

$$Y_{ij} = \gamma_{00} + u_{0j} + \varepsilon_{ij}$$

where = math score of student in school ; = grand mean of math score across all students and schools; = school-level random effect, showing how school 's mean differs from the grand mean; and = student-level residual, showing how student 's score differs from the expected mean score for their school.

In Model 1, student-level variables were entered. The model indicated to what extent differences in students' characteristics, attitudes, motivation, as well as home context are associated with math achievement. School level variables were entered in Model 2, a full random intercept model, where student-level and school-level variables were kept as fixed parameters. In Model 3, student ESCS was allowed to randomly vary across schools, while other variables were entered as fixed effects. In Model 4, cross-level and level 1 interactions were entered to assess the moderation effect, allowing the ESCS slope to vary across schools.

To correctly interpret cross-level interactions, student level variables are group centred, and school level variables and interaction variables are grand mean centred. The estimation used final student-level weights at Level 1 that were rescaled within schools for multi-level estimation and the Grade Nonresponse Adjusted School Base Weight as the level-2 sampling weight to account for school selection probabilities and adjust for school nonresponse (Mang et al., 2021). Replicate-weight variance estimation was not implemented in MLwiN, due to the substantial computational demands involved.

Following OECD (2023) guidelines, plausible values were treated as multiple imputations and all plausible values were included in the analysis and were combined using Rubin's rules (1987) to obtain unbiased estimates and correct standard errors:

where is the estimate from plausible value , is average within-imputation variance, is between-imputation variance, and is the number of plausible values.

Results

In the unconditional model presented in Table 3, intraclass correlation coefficients indicated that 73.2% of the variation in math scores lay within schools between students, 26.8% of the variation between schools. This indicates a moderate to high degree of clustering, suggesting that students within the same school are relatively similar to each other and that school-level factors are significant.

Comparing the models, the progressive reductions in −2*log-likelihood from Model 0 to model 2 indicate improved model fit. Student level variables entered in Model 1 account for 21.0% of the variance at the student level and 41.7% at the school level. School context variables included in Model 2 explain 16.7% of the variance at the school level and 2.4% at the student level.

(Table 3 about here)

In Model 2, except family support, all other student level variables included in the model demonstrate significant effects on student performance in mathematics, especially math efficacy ($z = 18.5$, $p < .001$), immigrant status ($z = 14.9$, $p < .001$), math anxiety ($z = 11.4$, $p < .001$), social connections ($z = 9.5$, $p < .001$), studying/homework frequency, cognitive activation, curiosity, ESCS, well-being, and math preference. Among school context variables, school mean ESCS ($z = 14.32$, $p < .001$) is the most significant predictor of student math performance, along with instructional leadership and number of math ECAs. The effects of teacher-related factors affecting school climate, school encouragement of parent involvement, proportion of all teachers fully certified, and teacher participation are also significant. Math

teacher training and negative school climate are not significant predictors. The effect of ESCS reduces from Model 1 to Model 2, but remains highly significant. Among these significant effects, several are negative. For example, higher well-being and higher social connections are associated with lower scores.

Model 3 are fitted to the same observations as Model 2, with the random ESCS slope structure added. Adding a random ESCS slope reduced −2*log-likelihood by 13.04, which significantly improved model fit ($p < .002$). In model 3, the random-slope variance for student ESCS was 167.72 (SE = 87.93), corresponding to a standard deviation of 12.95. This indicates appreciable between-school heterogeneity in the association between student ESCS and achievement, supporting examination of whether school-level characteristics account for variation in the ESCS effect. The positive covariance 183.05 (SE = 101.35) suggests that schools with higher math achievement at the ESCS reference value tend to have steeper positive ESCS gradients**.**

In Model 4, the effects of several interaction terms are significant, indicating that the socio-economic gradient changes as some student and school level factors change. Notably, ESCS × IMMIG is highly significant and positive, suggesting that the socio-economic differences in math scores are greater when the immigrant background is captured. Specifically, the ESCS slope more positive for students of first generation immigrant background. Whether the effect of a student's ESCS on math achievement depends on the socioeconomic composition of the school is tested by the cross-level interaction term ESCS x Mean_ESCS, which is insignificant**.** ESCS × ENCOURPG and ESCS × STUBEHA are also insignificant, implying the individual socio-economic differences in math scores are not different in schools where parent participation is promoted differently or students behave differently. The significant, negative interaction of ANXMAT and Mean_ESCS implies that school-average socioeconomic status

moderates the association between math anxiety and math achievement. Math anxiety tends to be more strongly negatively associated with achievement in higher-ESCS schools.

Discussion

The analytical framework adopted in this study incorporates a range of potential predictors of students' mathematics achievement in the UAE's private schools. The results reveal several noteworthy divergences and provide new insights into the ways in which socio-economic status interacts with other determinants of achievement. Elsewhere, emerging scholarship argues that socioeconomic status interacts with ethnicity, migration background, language, and culture (Hu et al., 2026; Kumar et al., 2026). The lack of understanding of such influences matters particularly for the UAE, where immigrant status, school socioeconomic composition, as well as language of instruction may operate differently from other international contexts.

Among all predictors, student characteristics, cognitive and motivational factors show the strongest associations with mathematics performance. Mathematics self-efficacy, immigration background, and mathematics anxiety emerged as key predictors. Home-context variables, including students' social relationships and family socio-economic status, also play important roles. The strongest positive relationship between mathematics self-efficacy and achievement, as well as the strong but negative relationship between mathematics anxiety and performance, are consistent with a substantial body of prior research (Güzeller et al., 2016; Zhang et al., 2019).

Findings related to immigration status require contextual interpretation. International research reports mixed evidence regarding achievement differences between immigrant and non-

immigrant students (Campos et al., 2025). For example, analyses of PISA 2012 data in the United States found no significant differences in mathematics performance between first-generation immigrant students and later-generation students once individual and school-level characteristics were considered (Pivovarova & Powers, 2019). In their analysis of an international sample, Karakus et al. (2022) showed perceived parental support, teacher enthusiasm, and the adaption of instruction contributed to academic performance of immigrant students, while bullying demonstrated a negative relationship. However, the UAE presents a unique case because it is a global expatriate hub with a large proportion of immigrant students enrolled in private schools. Many of these students come from relatively affluent families who can afford high-fee schooling, private tutoring, and enriched learning environments, contributing to higher average socio-economic status levels. The ESCS index indicates that students attending the UAE's private schools generally come from moderately advantaged socio-economic backgrounds compared with the OECD average.

Although the UAE does not represent a typical immigrant-receiving education system, socio-economic status appears to be particularly consequential for first-generation immigrant students' academic outcomes. As the individual socio-economic gradient does not change across schools that differ in school encouragement of parental involvement and the socioeconomic composition of the school, it may be influenced by other school characteristics that are not accounted for in the study or by individual characteristics such as immigrant background. This implies that on the one hand the UAE's private schools do differ in various ways, and on the other hand socio-economic status and migration background jointly influence academic achievement (Boman & Wiberg, 2024).

The significant influence of individual and school ESCS on students' math performance highlights the presence of socio-economic inequity in mathematics literacy performance. Socio-economic resources could provide educational advantages in accessing learning resources and opportunities either via or outside the school system, offering those more recent immigrant students a favourable level playing ground of the educational system (Li & Heath, 2016). Likewise, Barnard-Brak et al.'s (2018) analysis of PISA data shows that socio-economic status does not explain academic mathematics achievement, when accounting for differences in opportunity to learn. Instructional quality, socio-economic background, and achievement, therefore, need to be examined jointly rather than independently. Many studies conceptualize SES as a direct predictor of math performance, although research increasingly treats it as a complex social process operating through parental involvement, educational expectations, cultural capital, and school contexts (Liu et al., 2024; Pinquart & Ebeling, 2020). However, this study provides only limited evidence about these mechanisms. It is worth noticing that the two-level models test direct effects and moderation rather than mediation. Therefore, while the findings of this research support the importance of socio-economic context, further analysis is required to identify the family or school processes that transmit socio-economic advantage.

The negative impact of bullying observed in this study aligns with prior research showing that victimization can undermine students' academic engagement by reducing effort, weakening self-regulation, and increasing vulnerability to anxiety (Riffle et al., 2021). Competitive school environments may further increase students' exposure to bullying behaviours (Huang et al., 2024). Although a sense of school belonging can mitigate the negative effects of bullying, the findings of this study suggest a more complex relationship. In contrast to much of the existing literature, which typically reports positive relationships between social relationships, well-being, attitude toward school, and academic outcomes (Korpershoek et al., 2020; Lee, 2016),

this study finds negative correlations between these variables and mathematics achievement. One possible explanation could be that in high-performing school contexts students who achieve strong results in mathematics may devote substantial time to independent study and academic work, limiting their social and school engagement (Heller-Sahlgren, 2018). Consequently, the negative correlations identified in this study may not be interpreted as evidence that well-being or social relationships undermine achievement (Clarke, 2020). Rather, they may reflect compensatory social dynamics or different patterns of engagement across academic and social domains within demanding educational environments.

Similarly, the observed negative relationship between mathematics motivation and mathematics achievement is also likely to reflect compensatory behavioural patterns among lower-performing students. Increased motivation and effort could represent attempts by struggling students to improve performance rather than indicators of higher competence (Hu et al., 2026; Santana-Monagas et al., 2025). This phenomenon resembles the “motivation-achievement paradox” often observed in international large-scale assessments (He & Van de Vijver, 2016). Within the context of the UAE’s private schools, many of which operate in high-performing and academically demanding environments, high motivation may coexist with elevated anxiety levels that ultimately suppress achievement outcomes. Moreover, motivation itself may vary across ecological contexts and achievement levels (Saglam & Goktenturk, 2024). Niehues et al.’s (2020) analysis of Turkish students’ math achievement in PISA suggested that motivation is linked with socio-economic status, primarily through school environment and personal competence beliefs. Longitudinal research suggests that relationships between motivation and achievement may differ across within-person and between-person processes (Santana-Monagas et al., 2025). Supporting this interpretation, Lee (2016) found no direct relationship between student attitudes and mathematics achievement in PISA analyses after controlling for gender,

socio-economic status, and ability. Furthermore, it is also important to distinguish between intrinsic and extrinsic motivation, as these forms of motivation may influence academic performance in different ways within the UAE educational context (Yang et al., 2018).

At the school level, school-mean ESCS, instructional leadership, and the availability of mathematics-related extracurricular activities were significantly associated with students' mathematics achievement. These findings are consistent with school-effectiveness research, which emphasizes that school composition, leadership, and learning opportunities can influence achievement beyond students' individual characteristics (Nash, 2003; Sciffer et al., 2020). The significant cross-level interaction suggests that higher-ESCS school contexts may not help reduce students' math anxiety. High-ESCS schools provide greater academic opportunities but may also create stronger achievement norms and comparison pressures, making anxiety more consequential for students' ability to benefit from those environments (D'Agostino et al., 2022; Svraka & Ádám, 2024). Thus, attending a higher-ESCS school could be a double-edged sword for some students, which is consistent with the "big-fish-little-pond" theory (Dai & Rinn, 2008).

The negative coefficient for schools' encouragement of parental involvement should be interpreted cautiously. As parental involvement declines as children move into middle and high school (Koivuhovi et al., 2025), it may reflect compensatory practices whereby schools promote parental engagement in response to existing academic difficulties. More broadly, schools have several potentially important levers for improvement, but their influence remains embedded within wider cultural and institutional contexts. As Breton (2023) argues, school-based initiatives may have limited effects when they are not accompanied by broader cultural conditions that support achievement. Accordingly, effective improvement strategies should

combine strong instructional leadership and enriched mathematics opportunities with context-sensitive efforts to strengthen supportive norms for learning.

Beyond school-level factors, the strong PISA performance of UAE private schools may reflect broader system-level dynamics, including accountability mechanisms, benchmarking practices, and sustained policy pressure for improvement. Data from the Dubai Knowledge and Human Development Authority indicate that the performance gap between schools rated “Acceptable” and those rated “Outstanding” exceeds 60 points on the PISA scale (KHDA, 2023), highlighting substantial variation associated with school quality. Although the present study does not include school inspection ratings or curriculum categories, it is plausible that higher-rated schools enrol students from more advantaged socio-economic backgrounds, while lower-fee schools and certain curricula serve families with more modest financial resources.

In summary, the multilevel analysis indicates substantial between-school variation in mathematics achievement and demonstrates that student performance is shaped by interactions between individual characteristics and school contexts rather than remaining constant across educational environments. The results support the view that family and school environments can moderate the advantages or disadvantages associated with individual background characteristics. Therefore, the findings contribute not only to cognitive perspectives on mathematics learning but also to socio-cultural theories emphasizing that learning is embedded within specific social and institutional contexts.

(Table 4 about here)

Having shed light on some dynamic relationships between some student and school level factors, it should be noted that there may be reciprocal relationships between math achievement and anxiety (Pelegrina et al., 2026), self-efficacy, motivation, and other factors. Reciprocal

relationships could help us to better understand achievement as part of a developmental feedback system, rather than just as the dependent variable of a set of predictors, which this present study has attempted to achieve. More advanced statistical procedures or longitudinal designs, therefore, are needed to illustrate the likely reciprocal processes. Longitudinal research is particularly useful in examining how mathematics achievement and its determinants evolve over time, addressing the limitation of using cross-sectional data to draw causal conclusions regarding the relationships observed.

Several other limitations should also be acknowledged. First, the analysis relies exclusively on data from PISA 2022 and therefore does not incorporate additional classroom- or system-level variables that may influence math achievement. Second, although the analysis accounts for plausible values and sampling weights, replicate-weight variance estimation was not implemented. Consequently, the standard errors may not fully reflect uncertainty arising from PISA's complex sampling design and thus should be interpreted with caution. Third, the dichotomous coding of several constructs may have reduced statistical information and obscured meaningful variation within categories. In addition, qualitative and mixed-method approaches may provide deeper insights into the mechanisms through which classroom practices, school environments, and family contexts influence students' mathematics learning. Such research would contribute to a more comprehensive understanding of mathematics education in diverse educational settings and support the development of more effective policy and instructional strategies.

Although the present research concerns private schools, its findings also have implications for UAE public schools. Research evidence from Abu Dhabi indicates that students' engagement in public schools is likewise shaped by school environments (Yang et al., 2017). The strong

correlations of self-efficacy, anxiety, and cognitive activation with achievement therefore suggest that improvement strategies in public schools should extend beyond curriculum coverage to strengthen students' confidence, reduce anxiety, and provide demanding but supportive instruction. The contrasting gender patterns revealed by this study and the UAE PISA 2022 country report suggest that gender differences are context-dependent rather than fixed (OECD, 2023c; Thien, 2016), public-school policies should use school-level evidence to identify where boys require targeted academic and motivational support. More broadly, the substantial between-school variation found here points to the value of differentiated assistance for lower-performing public schools. The differences in school resources across public schools may be small. More focus should be placed on stronger instructional leadership, mathematics enrichment opportunities, and targeted support for students from less advantaged backgrounds.

Conclusions

This study addresses an important gap in the empirical literature on mathematics education in the UAE by applying a multilevel analytical framework to examine the determinants of student mathematics achievement in private schools. The findings demonstrate that mathematics achievement is shaped by a complex set of interacting influences across multiple levels of the educational system. These findings provide important insights for improving mathematics education not only in UAE private schools but also in public schools and other urban education systems seeking to balance high performance with educational equity. The evidence from this study suggests that isolated policy interventions are unlikely to produce substantial improvements. Integrated approaches addressing multiple dimensions of the learning environment are needed.

Given the strong relationships identified between self-efficacy, mathematics anxiety, and achievement, strengthening students' foundational skills at earlier stages of schooling should be prioritized. Schools may benefit from adopting evidence-based instructional practices that emphasize structured learning routines, explicit problem-solving strategies, and opportunities for guided practice. In addition, incorporating elements such as growth mindset development and metacognitive learning strategies into mathematics instruction may help improve students' confidence and resilience in tackling challenging mathematical tasks. Schools should also actively monitor students' well-being and levels of mathematics anxiety, as emotional and psychological factors can significantly influence learning outcomes. Creating positive behavioural and disciplinary climates that ensure students feel safe and supported in their learning environments is important, especially in schools serving disadvantaged populations.

A more nuanced understanding of the relationships between individual SES, school SES, and student math achievement has important implications for policy makers and families, with the potential to inform policy on school segregation as well as school-choice decisions. Addressing disparities related to socio-economic status remains a priority for education policy. Targeted support should be provided to low-performing schools and to disadvantaged students within schools. Education authorities may consider strengthening infrastructure, expanding teacher professional development opportunities, and promoting learner-centred pedagogical approaches in schools facing greater challenges. At the same time, policies that broaden access to high-performing schools, such as fee-support programmes for capable students from lower socio-economic backgrounds, may help reduce opportunity gaps and promote greater equity within the education system.

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Table 1

Profile of Private Schools and Students in the Sample

	Number	Percentage
Gender		
Male	5,245	49.2
Female	5,423	50.8
Immigrant status		
1st generation	5,077	47.6
Native or 2nd generation	5,591	52.4
	Mean	Standard Deviation
Index of economic, social and cultural status	.426	.656
Life satisfaction cross domains	.142	1.229
Proportion of all teachers fully certified	.274	.208
Student/teacher ratio	15.190	4.997

Table 2

Correlation Matrix

	MATHPREF	MATHMOT	RELATST	BULLIED	FEELSAFE	CURIOAGR	FAMSUP	DISCLIM	EXPOFA	MATHEFF	ANXMAT	ESCS	SOCONPA	SOCCON
MATHPREF	1.000	0.188	-0.014	-0.013	0.037	-0.024	-0.001	0.020	0.027	0.129	-0.140	0.037	0.019	0.011
MATHMOT	0.188	1.000	-0.043	0.035	-0.027	-0.052	-0.058	-0.036	-0.001	-0.016	0.003	-0.024	-0.045	0.000
RELATST	-0.014	-0.043	1.000	-0.232	0.311	0.263	0.259	0.258	0.145	0.198	-0.183	0.008	0.273	0.221
BULLIED	-0.013	0.035	-0.232	1.000	-0.235	-0.050	-0.155	-0.281	-0.070	-0.092	0.116	0.023	-0.223	-0.149
FEELSAFE	0.037	-0.027	0.311	-0.235	1.000	0.199	0.161	0.158	0.097	0.185	-0.130	0.050	0.193	0.115
CURIOAGR	-0.024	-0.052	0.263	-0.050	0.199	1.000	0.209	0.138	0.131	0.257	-0.142	0.037	0.156	0.089
FAMSUP	-0.001	-0.058	0.259	-0.155	0.161	0.209	1.000	0.181	0.104	0.202	-0.149	0.075	0.442	0.283
DISCLIM	0.020	-0.036	0.258	-0.281	0.158	0.138	0.181	1.000	0.052	0.171	-0.167	-0.011	0.179	0.112
EXPOFA	0.027	-0.001	0.145	-0.070	0.097	0.131	0.104	0.052	1.000	0.203	-0.042	0.047	0.084	0.057
MATHEFF	0.129	-0.016	0.198	-0.092	0.185	0.257	0.202	0.171	0.203	1.000	-0.408	0.161	0.177	0.098
ANXMAT	-0.140	0.003	-0.183	0.116	-0.130	-0.142	-0.149	-0.167	-0.042	-0.408	1.000	-0.065	-0.169	-0.172
ESCS	0.037	-0.024	0.008	0.023	0.050	0.037	0.075	-0.011	0.047	0.161	-0.065	1.000	0.072	-0.032
SOCONPA	0.019	-0.045	0.273	-0.223	0.193	0.156	0.442	0.179	0.084	0.177	-0.169	0.072	1.000	0.384
SOCCON	0.011	0.000	0.221	-0.149	0.115	0.089	0.283	0.112	0.057	0.098	-0.172	-0.032	0.384	1.000

Table 3

Two-level Models for Mathematics Achievement

	Model 0		Model 1		Model 2		Model 3		Model 4	
	Estimate	S.E.	Estimate	S.E.	Estimate	S.E.	Estimate	S.E.	Estimate	S.E.
Fixed Part										
cons	475.31	1.33	451.02	2.48	452.04	2.37	452.36	2.36	455.44	2.59
Female			-7.88***	2.13	-8.02***	2.12	-8.13***	2.12	-7.53***	2.11
1st generation			31.10***	2.10	31.12***	2.09	30.98***	2.09	23.44***	3.07
MATHPREF			20.48***	2.90	18.62***	2.81	18.53***	2.80	18.32***	2.80
Not every day			22.24***	2.36	18.55***	2.32	18.38***	2.31	18.27***	2.30
More motivated			-18.37***	5.24	-15.00**	5.09	-14.97**	5.09	-15.22**	5.05
MATHEFF			20.83***	1.10	19.65***	1.06	19.60***	1.05	19.46***	1.05
ANXMAT			-9.87***	0.89	-10.02***	0.88	-10.03***	0.87	-4.17**	1.33
BULLIED			-5.57***	0.97	-5.46***	0.95	-5.45***	0.95	-5.42***	0.95
FEELSAFE			5.27***	1.14	4.84***	1.12	4.86***	1.12	4.84***	1.13
CURIOAGR			6.98***	0.99	7.33***	0.95	7.39***	0.95	7.38***	0.95
LIFESAT			-6.93***	0.98	-6.44***	0.97	-6.45***	0.96	-6.48***	0.97
RELATST			4.69***	1.02	3.74***	0.99	3.77***	0.99	3.87***	0.99
DISCLIM			4.10***	1.07	3.73***	1.03	3.70***	1.02	3.85***	1.02
COGACMCO			-8.99***	1.08	-8.23***	1.07	-8.25***	1.07	-8.06***	1.07
EXPOFA			4.23***	1.04	4.11***	1.01	4.05***	1.02	3.95***	1.01
FAMSUP			-0.74	1.12	-0.82	1.09	-0.84	1.09	-0.76	1.08
SOCONPA			6.60***	1.03	5.60***	1.01	5.61***	1.01	5.48***	1.01
SOCCON			-12.39***	1.15	-10.73***	1.14	-10.75***	1.14	-10.68***	1.14
ICTRES			-3.95***	1.02	-5.85***	1.03	-5.59***	1.04	-5.19***	1.05
ESCS			22.49***	1.89	13.29***	1.86	13.92***	1.84	9.44***	2.47
PROATCE					19.05***	4.80	19.26***	4.80	18.42***	4.80
TCHPART					3.56***	0.91	3.49***	0.90	3.46***	0.90
ENCOURPG					-4.66***	1.16	-4.60***	1.16	-4.98***	1.27
INSTLEAD					-6.39***	1.25	-6.40***	1.25	-6.21***	1.25
MTTRAIN					3.49	2.46	3.38	2.46	2.95	2.45
NEGSCLIM					-3.10	1.61	-3.03	1.59	-2.15	2.08
TEACHBEHA					-5.08***	1.05	-5.08***	1.04	-5.01***	1.04
0 or 1 math ECA					-18.81***	3.74	-18.45***	3.72	-17.74***	3.67
No grouping					11.77***	3.22	11.77***	3.22	11.61***	3.24
Mean_ESCS					50.13***	3.50	50.07***	3.47	50.08***	3.92
ANXMAT x NEGSCLIM									1.68*	0.79
ANXMAT x Mean_ESCS									-10.49***	2.23
IMMIG x NEGSCLIM									-1.69	2.23
ESCS x ENCOURPG									1.08	1.45
ESCS x Mean_ESCS									-6.39	3.66
ESCS x IMMIG									14.50***	2.76
Random Part										
Level: School	4168.41	325.06	2428.61	223.68	2021.85	213.19	1991.38	214.92	1952.82	216.36
							167.72	87.93	124.60	85.66
							183.05	101.35	146.64	95.61
Level: Student	11378.36	689.76	8993.77	485.13	8780.12	473.27	8704.78	469.41	8673.00	469.40
-2*loglikelihood	127747.02		124237.70		123529.30		123516.26		123418.12	
Units: School	104		104		104		104		104	
Units: Student	10668		10667		10667		10667		10667	

Note: *** significant at 0.001, ** significant at 0.01, * significant at 0.05; two-tailed

Table 4

Summary of Key Predictors of Math Performance

	Direction of relationship
Student level	
Immigration background (reference 2nd generation)	Positive
Math self-efficacy	Positive
Math anxiety	Negative
Socio-economic status (ESCS)	Positive
Mathematics motivation	Negative
Bullying	Negative
School level	
School-mean ESCS	Positive
Instructional leadership	Negative
Availability of mathematics-related extracurricular activities (reference 2 or more math ECA)	Positive